Week 04: The Chinese Room and Its Replies

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

Searle's Chinese Room is the most discussed argument against strong AI. This week reconstructs the argument, examines the main replies and Searle's answers, and asks what the argument shows about present language models [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to reconstruct the Chinese Room argument and its later axiomatic form. They should be able to explain the systems, robot and brain simulator replies and Searle's answers, and to assess the luminous room objection. Finally, they should be able to state precisely which conclusion the argument supports.

Lecture

The argument

Searle imagined himself locked in a room with a large rulebook written in English [1]. Batches of Chinese characters are passed in, and by following the rules, which relate shapes to shapes, he passes back other Chinese characters. The rules are so good that people outside, who read the answers, cannot tell them from those of a native speaker. Yet Searle understands no Chinese: He manipulates symbols by their shapes alone. Since a computer running a program does nothing more than Searle does in the room, running a program cannot be sufficient for understanding. The target is strong AI, not the usefulness of AI.

A decade later, Searle presented the reasoning as an argument from axioms [2]. Programs are formal, or syntactic. Minds have mental contents, or semantics. Syntax by itself is neither constitutive of nor sufficient for semantics. Therefore programs are neither constitutive of nor sufficient for minds. The interactive lab of this week puts students inside the room and then asks them to test their position on these axioms.

The replies

Searle discussed several replies in the original article [1]. The systems reply grants that the man does not understand but holds that the whole system, man, rulebook and papers together, does. Searle answered that he could memorise the rules and do everything in his head and would still understand nothing. The robot reply proposes putting the program in a robot with cameras and motors, so that its symbols are connected to the world. Searle answered that the new inputs are just more symbols to him. The brain simulator reply proposes a program that simulates the firing of neurons in a Chinese speaker's brain. Searle answered that simulating the formal structure of neuron firings misses the brain's causal powers, and he illustrated the point with a system of water pipes and valves.

Churchland and Churchland argued that the third axiom is not self-evident but an empirical claim, and they offered an analogy [4]. A person waving a magnet in a dark room produces electromagnetic waves but no visible light; it would be a mistake to conclude that light is not electromagnetic radiation. Intuitions about slow and small cases can mislead about fast and large ones. Preston and Bishop collected later essays that continue the debate from many directions [3].

What the argument shows and what it does not

Precision about the conclusion matters. The argument claims that running a program is not sufficient for understanding. It does not claim that machines cannot understand, since Searle held that a machine with causal powers equivalent to those of brains could think. It also leaves open what those causal powers are, which critics regard as a weakness. The argument relies on an intuition, that the man in the room does not understand, and on a step from the man to the system. Most replies attack that step or the reliability of the intuition, as Figure 4.1 shows. Harnad read the room as an illustration of a more specific problem, namely how symbols can be connected to what they mean, which is the topic of Week 5 [5].

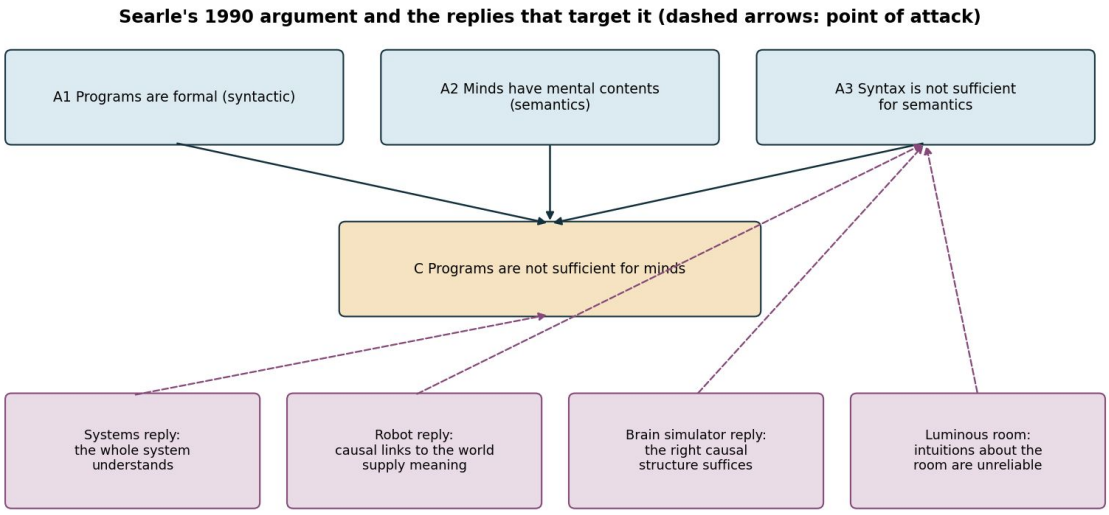


Figure 4.1. Searle's axiomatic argument and the points at which four replies attack it [2, 4].

The room and language models

Large language models are trained on text alone and produce fluent text by transforming symbols, so the argument seems to apply to them directly. There are also differences. No person wrote their rules; the rules are learned statistical regularities distributed over billions of parameters. Defenders of the systems reply argue that whatever understanding exists would belong to the trained system as a whole, not to any component. Critics answer that learning the rules does not change their syntactic character. The Colab notebook builds two rooms and compares how they handle new questions. One room only stores question-answer pairs, and the other composes answers from a small model of a world. Both only manipulate symbols, but they differ in how they generalise, which is one way of making the debate empirical.

> ***Pause and reflect.*** *In the interactive lab, did following the rulebook feel like understanding? Does your answer tell you anything about a system that follows billions of rules at high speed?*

Hands-on components

Interactive lab: Part A places you inside the room: notes arrive in Chinese, and a rulebook of shapes tells you which note to pass back. The outside view can be revealed afterwards. Part B tests your position on Searle's axioms, and Part C matches the replies to the step they attack [1, 2]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-04/lab.html

Colab notebook: It builds two rooms for a small symbolic world. One stores question-answer pairs, like a pure rulebook, and the other composes answers from a model of the world. It measures how each handles questions it has never seen and computes the size of a complete rulebook [1, 6]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-04/NB04_chinese_room.ipynb

Reflection prompts

1. Which reply do you find strongest, and what would Searle say in response? Write the exchange as a short dialogue.
2. Does the difference between the two rooms in the Colab notebook matter for the argument? Why or why not?
3. Apply the argument to a language model assistant you use. Which premise would its defenders reject?

Weekly task and submission

What to do. Write an argument analysis of the Chinese Room. First, reconstruct the argument in numbered form, either as the original story or as Searle's later axioms. Second, evaluate two replies: the systems reply and the luminous room objection of Churchland and Churchland. For

each reply, name the premise it attacks. Third, state whether the argument applies to present language models, and why.

Where to look. Lecture: sections "The argument", "The replies" and "The room and language models". Interactive lab: Part A, where you follow the rulebook yourself, Part B for Searle's axioms and Part C for the step that each reply attacks. Colab notebook: Sections 2 and 3, which compare a pure rulebook (Room 1) with a room that composes answers from a world model (Room 2). Exercise 1 measures how each room handles new questions.

Example. The robot reply, which the task does not ask about, claims that sensors and motors would connect the symbols to what they mean. It therefore attacks the step from symbol manipulation to a lack of meaning. Searle answered that the new inputs are just more symbols to him.

What to submit. About 600 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Searle, J. R. (1980). Minds, brains, and programs. Behavioral and Brain Sciences, 3(3), 417-424. https://doi.org/10.1017/S0140525X00005756

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[3] Preston, J., & Bishop, M. (Eds.) (2002). Views into the Chinese Room: New Essays on Searle and Artificial Intelligence. Clarendon Press.

[4] Churchland, P. M., & Churchland, P. S. (1990). Could a machine think?. Scientific American, 262(1), 32-37. https://doi.org/10.1038/scientificamerican0190-32

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[6] Block, N. (1981). Psychologism and behaviorism. The Philosophical Review, 90(1), 5-43. https://doi.org/10.2307/2184371